

AI-Powered Smart Traffic Management Report

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Executive Summary

This report presents the results of an AI-powered traffic analytics system built using YOLOv8, OpenCV, Machine Learning, and Streamlit. The system detects vehicles, analyzes traffic density, predicts congestion levels, and generates insights for traffic monitoring.

Traffic Statistics

Total Vehicles Detected: 6407

Average Vehicles Per Frame: 12.51

Maximum Vehicles Observed: 22

Vehicle Distribution

Cars: 5667

Motorcycles: 0

Buses: 390

Trucks: 350

Traffic Density Analysis

Low: 509

Medium: 3

Machine Learning Model Comparison

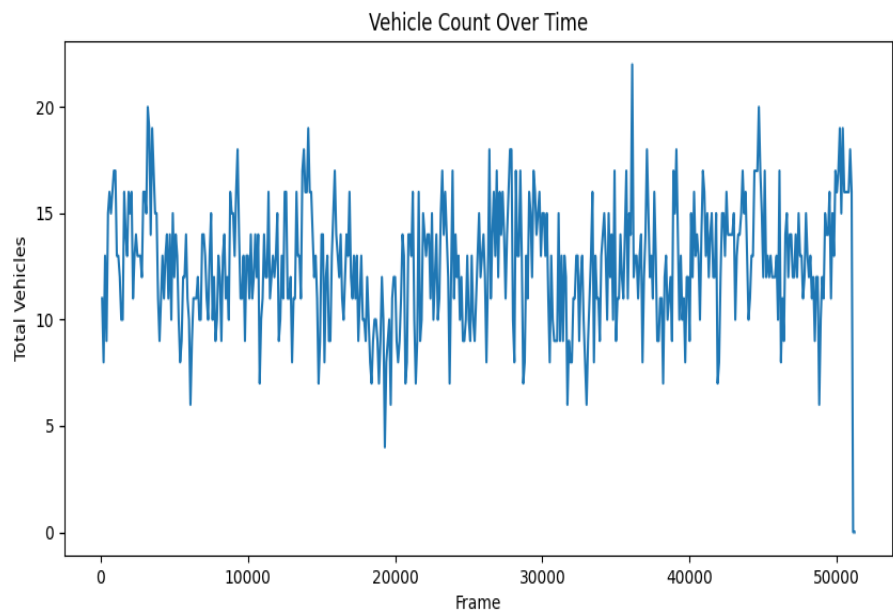
Random Forest : 99.03%

Decision Tree : 100.0%

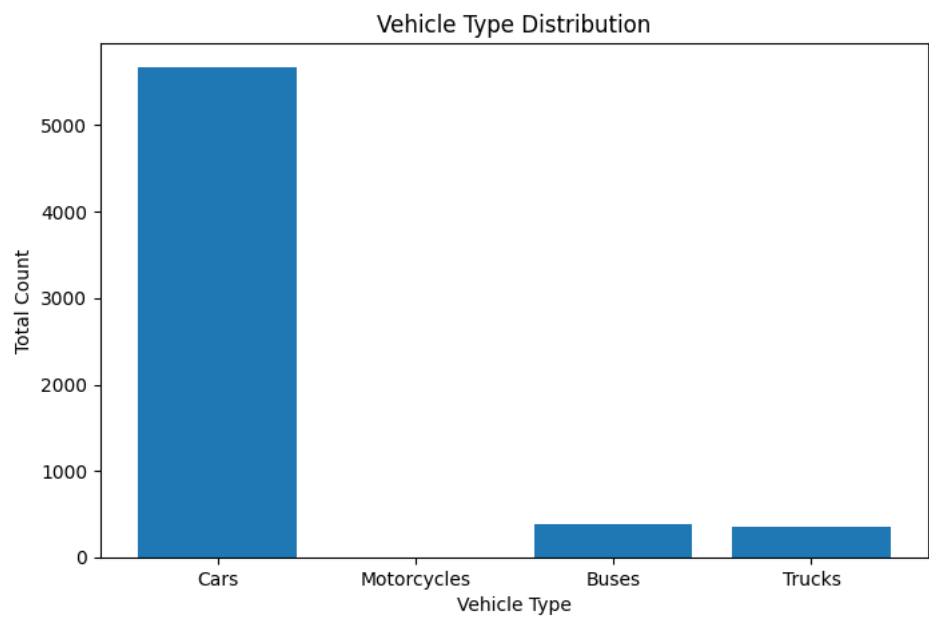
Logistic Regression : 99.03%

Analytics Visualizations

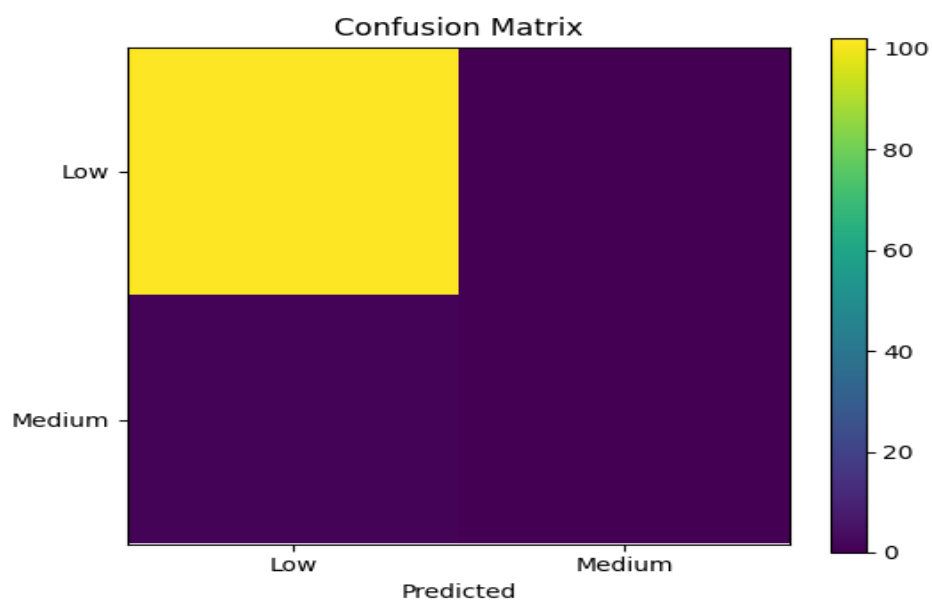
Traffic Trend



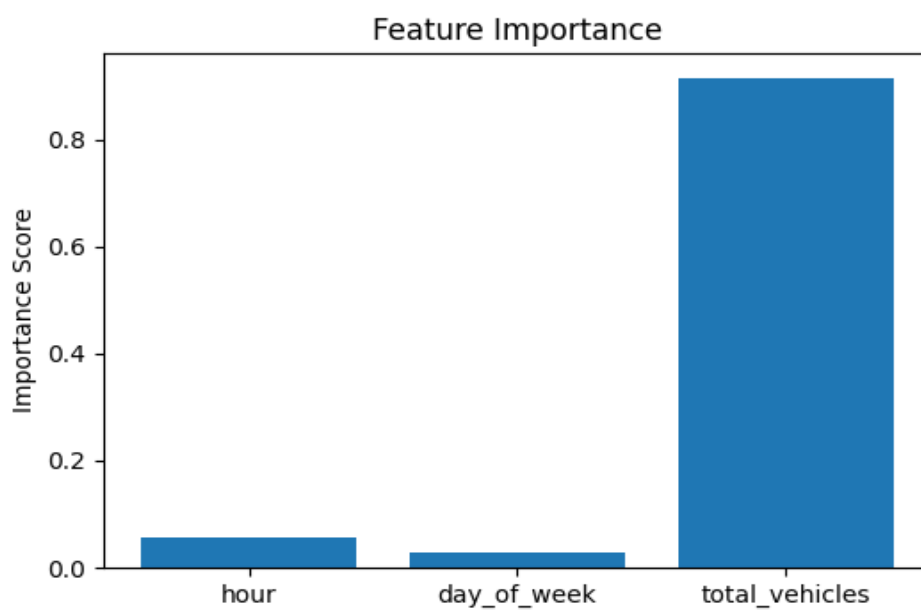
Vehicle Distribution



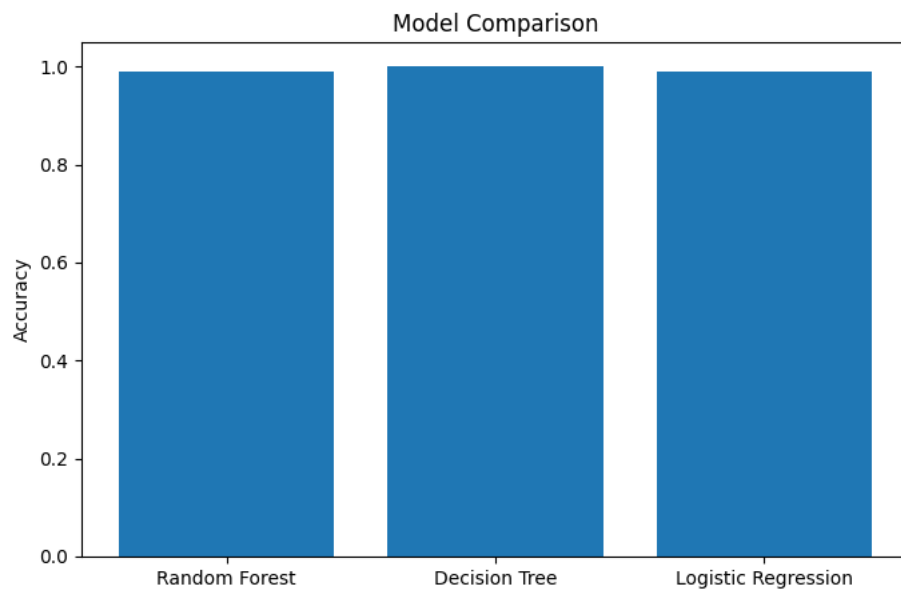
Confusion Matrix



Feature Importance Analysis



Model Performance Comparison



Traffic Feature Correlation Analysis

